

The Linear Representation Hypothesis and the Geometry of Large Language Models

Week 7: July 12th to July 18th

Questions

motivated by a central phenomenon:

“Linear Representation Hypothesis” : high-level concepts are represented linearly in the representation space of a model

- **What does the “linear representation hypothesis” actually mean?**
 - it's unclear what exactly "linear representation" entails—does it mean subspaces, measurements, or interventions?
 - **Subspace (Geometric/Structural view):**
 - Each high-level concept (e.g., gender, royalty) is treated as a **1-dimensional subspace** in the representation space.
 - Differences between related words (e.g., *woman* – *man* and *queen* – *king*) lie along the same subspace direction.
 - The concept is defined by this **shared direction** (axis) — such as a "Male/Female" concept axis.
 - **Measurement (Probing view):**
 - The value of a concept (e.g., whether a sentence is in French) can be **measured by a linear function (probe)**.
 - The probability of a concept value is assumed to be **logit-linear** in the representation of the input.
 - The linear map (probe) itself is the representation of the concept — it lets us *detect or read off* the concept.
 - **Intervention (Control/Steering view):**
 - The value of a concept can be **actively changed** without affecting unrelated concepts.
 - Done by adding a **steering vector** (direction) — e.g., pushing the representation along the "English → French" axis to flip the language.
 - The added vector is taken to be the **representation of the concept** in this view, enabling direct *control*.
 - **How can we relate these linear representations to interpretation and control of models?**
 - Can we use them to steer models (e.g., change outputs along some concept axis) or to measure properties (e.g., gender, language) with a probe?
 - **Which inner product should be used to make sense of geometric notions like similarity, projection, and directions in these spaces?**
 - it is not clear which inner product is appropriate for understanding model representations

Contributions

- **Formalizing subspace notion of linear representations via counterfactual pairs**
 - provide two formalizations:
 - In the **unembedding space (output word space)**, tied to *measurement* (probing concepts).
 - In the **embedding space (input context space)**, tied to *intervention* (steering outputs).
- **Introducing the causal inner product**
 - define a special inner product where causally separable concepts (that can vary freely from each other, like gender vs. language) are represented as *orthogonal directions*.
 - This inner product also *unifies embedding and unembedding representations*, ensuring both spaces can be consistently used.
- **Empirical validation and practical tools**
 - use LLaMA-2 experiments to show that:
 - Concepts indeed act as linear directions.
 - The causal inner product respects semantic structure.
 - Subspace-based representations can be used for **constructing probes and intervention vectors** reliably.

Background of language models

A language model (LM) processes a **context** x (e.g., prior text) and generates output tokens **one by one** (word by word or token by token)

- To compute the probability of the next token, the LM:
 - **Maps the context** x into a vector $\lambda(x)$ in a **representation space** $\Lambda \subseteq \mathbb{R}^d$ (**embedding vectors**).
 - **Represents each possible token** y as an **unembedding vector** $\gamma(y)$ in a **separate representation space** $\Gamma \subseteq \mathbb{R}^d$.
- The probability of predicting token y given context x is proportional to:

$$\mathbb{P}(y | x) \propto \exp(\lambda(x)^\top \gamma(y)).$$

meaning prediction depends on the **dot product (similarity)** between the context's embedding and the token's unembedding.

- **Two spaces: Embedding space (Λ)** for encoding the context.
Unembedding space (Γ) for scoring each candidate token

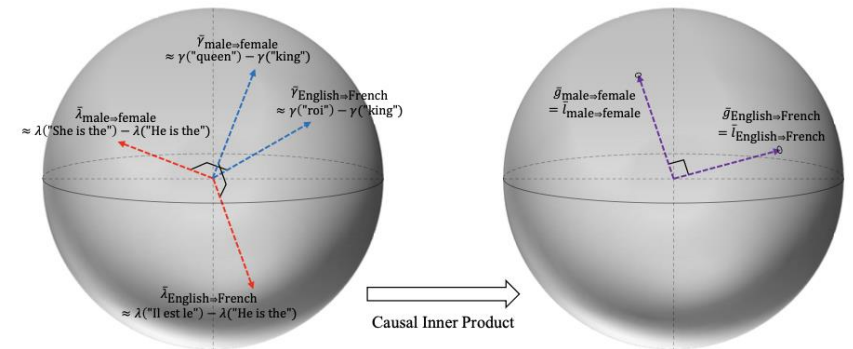


Figure 1. The geometry of linear representations can be understood in terms of a *causal inner product* that respects the semantic structure of concepts. In a language model, each concept has two separate linear representations, $\tilde{\lambda}$ (red) in the embedding (input context) space and $\tilde{\gamma}$ (blue) in the unembedding (output word) space, as drawn on the left. The causal inner product induces a linear transformation for the representation spaces such that the transformed linear representations coincide (purple), as drawn on the right. In this unified representation space, causally separable concepts are represented by orthogonal vectors.

The Linear Representation Hypothesis

formalizing the subspace notion of linear representation for two spaces, then tie the subspace notions to the measurement and intervention notions

- **Concepts:**

defined as any factor of variation that can be changed independently (e.g., switching language from French to English or gender from male to female without changing the language it is written in)

- **Concept Variables (W):**

- Each concept is treated as a **latent variable** caused by the context X and acting as a cause of the output Y
- An ordering is imposed (e.g., female $\rightarrow$ male) so that the direction/sign of the representation has meaning
- Focus on binary concepts (e.g., male $\rightarrow$ female) for simplicity

- **Counterfactual Outputs:**

- Each concept W defines a set of counterfactual outputs $\{Y(W=w)\}$ differing only by the value of W :
 - E.g. for concept **male** $\rightarrow$ **female**, $(Y(0), Y(1))$ is a *random element of the set* (“man”, “woman”), (“king”, “queen”), etc.

- **Deterministic Concept Values:**

- The value of a concept can be **directly inferred from the sampled output** (e.g., the token “king” implies $W=0$).

- **Relationships Between Concepts (Causal Separability):**

- Two concepts W and Z are **causally separable** if their counterfactual outputs can **vary freely and independently**
 - (e.g., English $\rightarrow$ French and male $\rightarrow$ female can vary freely, but English $\rightarrow$ French and English $\rightarrow$ Russian cannot).
- Causally separable concepts can be combined without conflict

- **Unembedding Representations and Measurement:**

A concept can be represented linearly in either embedding or unembedding space

- **Definition:** (concepts are represented as directions; direction is unique)

- A concept W is represented in the unembedding space if:

$$\gamma(Y(1)) - \gamma(Y(0)) \in \text{Cone}(\bar{\gamma}_W) \text{ almost surely.}$$

- This means the difference between counterfactual outputs (e.g., “king” vs “queen”) must align (up to positive scaling) with a concept direction γ^*W
- Using a **cone** (instead of a subspace) ensures we account for directionality (sign), since the axis for “king vs queen” should be the opposite of “man vs woman.”

The Linear Representation Hypothesis

- **Unembedding Representations and Measurement:**

- Connection between *Measurements (probing)* and *unembedding representations*:

$$\text{logit } \mathbb{P}(Y = Y(1) \mid Y \in \{Y(0), Y(1)\}, \lambda) = \alpha \lambda^\top \bar{\gamma}_W, \quad \leftarrow \text{probe-adjusted, logit-scaled concept activation}$$

where $\alpha > 0$ (a.s.) is a function of $\{Y(0), Y(1)\}$.

- The **direction** $\bar{\gamma}^\top W$ stays fixed across all pairs for a concept
- $\alpha > 0$ is a random scalar depending on the particular counterfactual pair:
 - **It adapts** to the specific **instantiation** of the concept through a **pair of counterfactual tokens**; adjusts the scale of that direction **as it's reflected in the embedding space (lens specific scaling)**
 - E.g.: {"king", "queen"}, {"roi", "reine"}, {"man", "woman"}, {"he", "she"}:
 - These pairs all express the **same underlying concept** (e.g., gender or royalty), but they differ in:
 - **Language** (English vs. French)
 - **Register** (formal vs. casual)
 - **Semantic nuance** (e.g., "man/woman" is broader than "king/queen")
 - directly grounded by how lm predict outcomes
 - Difference from Probes:
 - **not capture correlated off-target concepts** from a regression fit (like a probe might)
 - avoids spurious correlations: e.g. if French text were disproportionately about men, a probe could learn this information, but unembedding representation won't

- **Embedding Representations and Intervention:**

- **Definition:** $\bar{\lambda}_W$ is an *embedding representation* of a concept W if have $\lambda_1 - \lambda_0 \in \text{Cone}(\bar{\lambda}_W)$ for any context embeddings $\lambda_0, \lambda_1 \in \Lambda$ that satisfy

$$\frac{\mathbb{P}(W = 1 \mid \lambda_1)}{\mathbb{P}(W = 1 \mid \lambda_0)} > 1 \quad \text{and} \quad \frac{\mathbb{P}(W, Z \mid \lambda_1)}{\mathbb{P}(W, Z \mid \lambda_0)} = \frac{\mathbb{P}(W \mid \lambda_1)}{\mathbb{P}(W \mid \lambda_0)},$$

for each concept Z that is causally separable with W .

- $\lambda_1 - \lambda_0$, lies within the cone defined by $\bar{\lambda}^\top W$, meaning this difference points in the direction that represents W
 - The ratio > 1 ensures the shift actually moves in the positive direction of the concept
 - The representation doesn't affect **off-target concepts** Z ; shift about W won't leak into other concepts

The Linear Representation Hypothesis

- **Embedding Representations and Intervention:**

- **Connection between *Interventions (steering)* and *embedding representations*:**

- **Unembedding-embedding relationship:**

- Links the embedding representation $\bar{\lambda}_W$ to the unembedding representation $\bar{\gamma}_W$: $\bar{\lambda}_W^\top \bar{\gamma}_W > 0$ *and* $\bar{\lambda}_W^\top \bar{\gamma}_Z = 0$.
 - They must align for W
 - They must be orthogonal for unrelated, causally separable concepts Z
 - Why need off-target concepts to form lemma: If you're allowed to move in **any direction** that increases gender classification performance, you'll likely choose a **shorter path that also increases formality**. But now, when you test for formality — that direction gives spurious predictions.

- **Intervention representation:** how embedding representations govern **controlled interventions**

Theorem 2.5 (Intervention Representation). *Let $\bar{\lambda}_W$ be the embedding representation of a concept W . Then, for any concept Z that is causally separable with W ,*

$$\mathbb{P}(Y = Y(W, 1) \mid Y \in \{Y(W, 0), Y(W, 1)\}, \lambda + c\bar{\lambda}_W)$$

is constant in $c \in \mathbb{R}$, and

$$\mathbb{P}(Y = Y(1, Z) \mid Y \in \{Y(0, Z), Y(1, Z)\}, \lambda + c\bar{\lambda}_W)$$

is increasing in $c \in \mathbb{R}$.

In words: adding $\bar{\lambda}_W$ to the language model representation of the context changes the probability of the target concept (W), but not the probability of off-target concepts (Z).

- Embedding representations define the directions along which you can “steer” or manipulate a concept W in the model (like pushing a sentence’s meaning toward French or male), without unintentionally shifting unrelated concepts.
 - This gives a **mechanism for controlled interventions**, essential for understanding or editing model behavior.

Inner Product for Language Model Representations

earlier sections already defined linear directions (like the embedding representation of concepts, counterfactual movement, intervention directions, etc.). But those definitions **only constrain directions, not how we measure similarity or project onto them**.

Inner Product for Language Model Representations

- **Unidentifiability of the inner product**

- **Consider transforming embedding and unembedding space:** $g(y) \leftarrow A\gamma(y) + \beta, \quad l(x) \leftarrow A^{-\top}\lambda(x),$
- We see the transformation preserves the softmax distribution $P(y|x) \propto \exp(l(x)^\top g(y))$
 - only depends on the **dot product between $l(x)$ and $g(y)$** , not on their absolute positions or specific coordinate basis; never constrains the representation space itself to a fixed orientation or scale
 - But the **inner product between two concept directions** (like $\langle \gamma^W, \gamma^Z \rangle$) is no longer semantically fixed because the transformation A can arbitrarily rotate, stretch, or skew the space
 - given any fixed inner product, $\langle \bar{\gamma}_W, \bar{\gamma}_Z \rangle \neq \langle A\bar{\gamma}_W, A\bar{\gamma}_Z \rangle$, algebraic manipulations based on, e.g., the Euclidean inner product, won't be semantically meaningful

- **An Explicit Form for Causal Inner Product**

- **Motivation:**

- **Infeasible** from solving for a matrix that makes each pair of causally separable directions orthogonal; hard to compute

- **Key insight of simplified approach:**

- The value of a concept W for a randomly chosen word gives almost no information about another causally separable concept $Z \Rightarrow$ assumption: projections of random unembedding vectors onto the embedding directions for W and Z are independent

Assumption 3.3,

$$M^{-1} = GG^\top \text{ and } G^\top \text{Cov}(\gamma)^{-1}G = D, \quad (3.2)$$

for some diagonal matrix D with positive entries, where γ is the unembedding vector of a word sampled uniformly at random from the vocabulary.

$G = [\bar{\gamma}_{W_1}, \dots, \bar{\gamma}_{W_d}]$ forms a basis for the unembedding space.

D is a diagonal matrix (free parameter) with d degrees of freedom, allowing flexibility in defining the inner product.

- As we have: $\text{Cov}(\bar{\lambda}_W^\top \gamma, \bar{\lambda}_Z^\top \gamma) = 0$. $\langle \bar{\gamma}_W, \bar{\gamma}_Z \rangle_C = \bar{\gamma}_W^\top M \bar{\gamma}_Z = \bar{\gamma}_W^\top \text{Cov}(\gamma)^{-1} \bar{\gamma}_Z$, ensures orthogonality

- **Explicit form:** invariant under affine transformations

$$\langle \bar{\gamma}, \bar{\gamma}' \rangle_C := \bar{\gamma}^\top \text{Cov}(\gamma)^{-1} \bar{\gamma}', \quad \forall \bar{\gamma}, \bar{\gamma}' \in \bar{\Gamma}.$$

Experiments

- **Fig 2. Concepts as directions (in unembedding space)**
- Verify that **concepts (like Male vs Female, or English vs French)** can indeed be represented as *directions* (vectors) in unembedding space:
 - If **counterfactual pairs** (e.g., $Y(0)$ vs $Y(1)$) for a concept line up along the same axis (positive vs negative direction).
 - If **off-target pairs** (concepts that are causally separable from the target) are close to orthogonal.
- **Links to theory:**
 - **basic claim** that concepts are linear directions
 - directional representations can separate factual vs counterfactual outcomes.

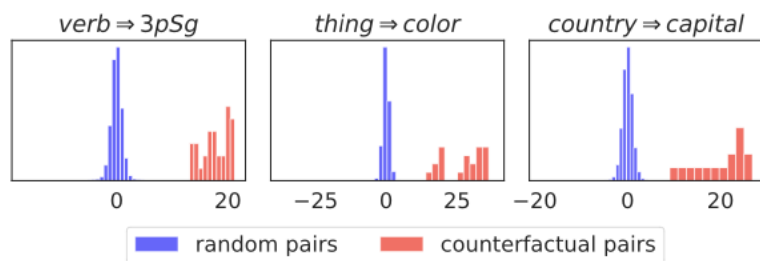


Figure 2. Projecting counterfactual pairs onto their corresponding concept direction shows a strong right skew, as we expect if the linear representation hypothesis holds. The projections of the counterfactual pairs, $\langle \tilde{\gamma}_{W,(-i)}, \gamma(y_i(1)) - \gamma(y_i(0)) \rangle_C$, are shown in red. For reference, we also project the differences between 100K randomly sampled word pairs onto the estimated concept direction, as shown in blue. See Table 2 for details about each concept W (the title of each plot).

- **Fig 3 . Inner product respects causal separability**
- Test if the **causal inner product** (from Section 3 and Theorem 3.4) behaves as designed:
 - Inner product between a concept's own direction and itself is **1 (normalized)**.
 - Inner product between **causally separable concepts** (like Male/Female vs English/French) is **0 (orthogonal)**.
- **Links to theory:**
 - Directly tests **Definition 3.1 (Causal Inner Product)** and its requirement for orthogonality.
 - Confirms that the inner product derived via covariance normalization actually preserves causal separability.

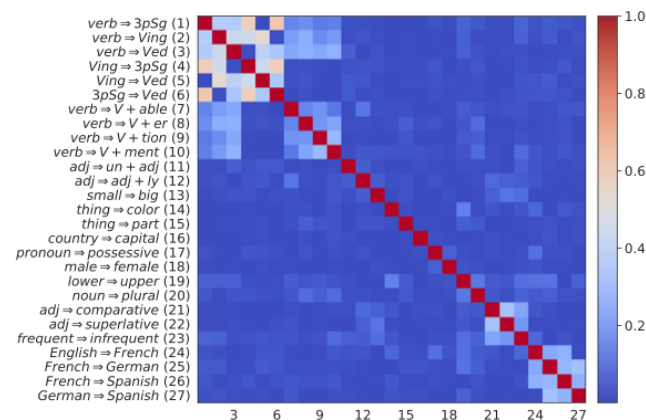


Figure 3. Causally separable concepts are represented approximately orthogonally under the estimated causal inner product based on (3.3). The heatmap shows $|\langle \tilde{\gamma}_W, \tilde{\gamma}_Z \rangle_C|$ for the estimated unembedding representations of each concept pair (W, Z) . The detail for each concept is given in Table 2.

Experiments

- **Fig 4 . Concept directions act as linear probes**
- Do **concept directions** work as **linear probes** for predicting the presence of a concept from the context?
- In other words: Can we take the direction (from the unembedding space) and use it to *linearly* predict factual vs counterfactual labels (like “is this male or female?”) in the context embeddings?
- **Links to theory:**
 - This connects strongly to **Theorem 2.2 (Measurement Representation)** and **Definition 2.3 (Embedding Representation)**.
 - It checks if the direction, when applied to the **embedding space ($\lambda(x)$)**, actually functions as a linear classifier (as predicted by theory).

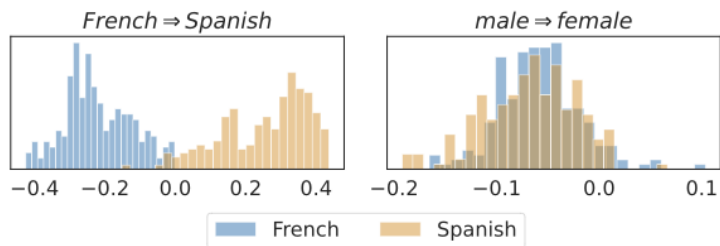


Figure 4. The subspace representation $\bar{\gamma}_W$ acts as a linear probe for W . The histograms show $\bar{\gamma}_W^\top \lambda(x_j^{\text{fr}})$ vs. $\bar{\gamma}_W^\top \lambda(x_j^{\text{es}})$ (left) and $\bar{\gamma}_Z^\top \lambda(x_j^{\text{fr}})$ vs. $\bar{\gamma}_Z^\top \lambda(x_j^{\text{es}})$ (right) for $W = \text{French} \Rightarrow \text{Spanish}$ and $Z = \text{male} \Rightarrow \text{female}$, where $\{x_j^{\text{fr}}\}$ and $\{x_j^{\text{es}}\}$ are random contexts from French and Spanish Wikipedia, respectively. We also see that $\bar{\gamma}_Z$ does *not* act as a linear probe for W , as expected.

French $\Rightarrow$ Spanish. To construct a dataset of French and Spanish contexts, we sample contexts of random lengths from Wikipedia pages in each language. Note that these are **not counterfactual pairs**. Following Theorem 2.2, we expect $\bar{\gamma}_W^\top \lambda(x_j^{\text{fr}}) < 0$ and $\bar{\gamma}_W^\top \lambda(x_j^{\text{es}}) > 0$. Figure 4 con-

an off-target concept $Z = \text{male} \Rightarrow \text{female}$ does not have any predictive power for this task (right). Appendix D.3

(“king”) decreases. We show the top-5 most probable words over the entire vocabulary following the intervention (4.2) in the $W = \text{male} \Rightarrow \text{female}$ direction, i.e., $\lambda_{W,\alpha}(x) = \lambda(x) + \alpha \bar{\gamma}_W$, for $\alpha \in \{0, 0.1, 0.2, 0.3, 0.4\}$. The original context $x = \text{“Long live the ”}$ is a sentence fragment that ends with the word $Y(W = 0)$ (“king”). The most likely words reflect the concept, with “queen” being top-1. In Appendix D.5, we provide more examples.

Rank	$\alpha = 0$	0.1	0.2	0.3	0.4
1	king	Queen	queen	queen	queen
2	King	queen	Queen	Queen	Queen
3	Queen	king	-	lady	lady
4	queen	King	lady	woman	woman
5	-	-	king	women	women

- **Fig 5 . Concept directions map to intervention representations**
- They test if adding the **concept’s direction** (in the embedding space) acts like an **intervention**:
 - It should **shift the probability of the target concept (W)**,
 - While **not affecting off-target concepts (Z)**.
- **Links to theory:**
 - Directly tests **Theorem 2.5 (Intervention Representation)**.
 - Validates that the embedding directions aren’t just linear probes but can act like *causal interventions*.

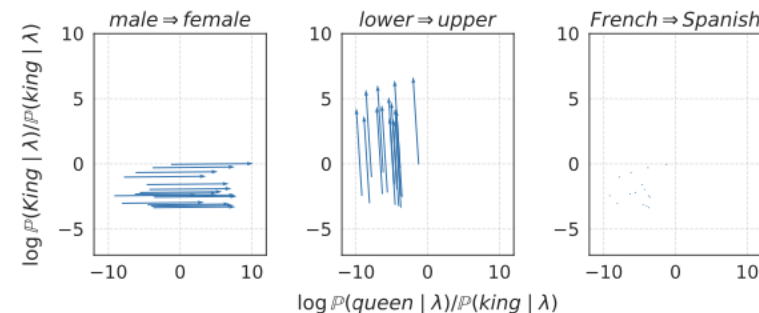


Figure 5. Adding $\alpha \bar{\gamma}_C$ to λ changes the target concept C without changing off-target concepts. The plots illustrate change in $\log(\mathbb{P}(\text{“queen”} | x)/\mathbb{P}(\text{“king”} | x))$ and $\log(\mathbb{P}(\text{“King”} | x)/\mathbb{P}(\text{“king”} | x))$, after changing $\lambda(x_j)$ to $\lambda_{C,\alpha}(x_j)$ as α increases from 0 to 0.4, for $C = \text{male} \Rightarrow \text{female}$ (left), $\text{lower} \Rightarrow \text{upper}$ (center), $\text{French} \Rightarrow \text{Spanish}$ (right). The two ends of the arrow are $\lambda(x_j)$ and $\lambda_{C,0.4}(x_j)$, respectively. Each context x_j is presented in Table 4.

discussion

- Power causal reasoning under nonstationary time-evolving or domain-shifting environments caused by changing treatment mechanisms:
 - As mentioned in 'Causal Modeling of Policy Interventions From Treatment–Outcome Sequences' paper, the policy intervention in real world is hard to observe, directional representation of $\mathbf{I}_m$ might help get free the need of prior knowledge of policy treatment from past:
 - Getting know which concept are changing?
 - Each timepoint t has an embedding λ_t . The change $\lambda_{t+1} - \lambda_t$ (or some counterfactual pair of policy concepts ?) tells you which concept was intervened on
 - If this direction lies in the cone of concept W , then this timepoint reflects a semantic shift in W
 - Domain shift question:
 - In COSTAR paper it proposed data augmentation way for cold start issue:
 - but the model relied on synthesized data, ood generalization depends heavily on how realistic this augmentation is.
 - If domain shift is caused by non-observed latent variables or policy-dependent selection bias, this augmentation may not fully cover the shifted state space.
 - It may also assume stationarity in the augmentation process
 - In domain adaptation or cold start, we often lack labeled data in the target domain. But if you know:
 - The direction of shift in latent concepts (e.g., topic shift, style change, formality), and
 - If we have these concepts are linearly encoded (as shown by cone subspace structure), Then:
 - You can project the domain shift vector into known concept subspaces, and measure how aligned it is with known semantics (how condition of samples changed due to past treatment decision)
 - This gives you:
 - A causal handle on why generalization fails
 - A way to extrapolate from source to target by keeping directions aligned
 - Counterfactual generation from concept intervention: from $\mathbf{I}_m$ geometry
 - Identify the concept vector (e.g., "gender") via $\gamma^T W$
 - Adjust the embedding of the input $\lambda(x)$ by adding or subtracting that direction
 - Feed it into the model to simulate the counterfactual outcome